

The Impact of Noise Distraction on Mind-wandering During Problem-solving

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Research Motivation

It has always been a concern to increase productivity and creativity, especially during problem-solving. However, environmental distractions such as noise, can be detrimental, disrupting the focus and attention of individuals during their daily tasks. Mind-wandering can be considered beneficial while solving these tasks. Yamaoka and Yukawa (2020g) found a positive relationship between mind-wandering and the ability of divergent thinking. This suggests that mind-wandering can help people generate ideas by exploring many solutions throughout the problem-solving. Although, as mentioned earlier, this advantageous effect can be impacted by environmental factors; leading to attenuated productivity among students, causing drawbacks to their efficiency and decision-making.

Moreover, we focus on the specific impact of sound distractions, in particular environmental noise, on the wandering of the mind and the cognitive performances of students during problem-solving tasks, and more precisely on route planning. Potential findings from our research can help various organizations change their strategies concerning the workplace to minimize environmental distractions which might enhance productivity.

Furthermore, from a practical point of view, the quality of the learning environment has a major impact on students' academic results. Former studies such as those of Shield & Dockrell (2008) and Hygge et al. (2002b) have shown that environmental noise can lead to decreased concentration and impaired cognitive performance among students. By understanding precisely how noise disrupts the route planning process and whether it affects mind wandering among students, we could identify effective ways to design learning environments that contribute positively to concentration and academic performance.

In addition, from a scientific point of view, this study will allow us to explore more in-depth the underlying mechanisms involved in the interactions between noise, cognitive functioning, and the behavior of individuals. A study conducted by Beaman et al. (2014b) has shown the effects of noise on concentration capacity and cognitive load in individuals. By examining how noise specifically affects students' ability to concentrate and plan routes, we could better understand the neurocognitive processes involved. This knowledge contributes to our understanding of the effects of noise on human cognition in an educational context, paving the way for new research and interventions to optimize learning environments.

Research Question

How do noise distractions, modulate the occurrence and impact of the mind wandering during route-finding tasks in university students?

Background

To begin with, in the scientific literature, the effect of noise on cognition and mind wandering has been studied in various contexts, ranging from work environments to classrooms to public spaces. These studies found that noise, particularly constant or intermittent background noise, can have significant impacts on concentration, memory, decision-making, and other aspects of cognitive performance. While a previously conducted study by Kane et al. (2007c) investigated how mind-wandering and working memory capacity are related, it also pointed out the significance of the influence of environmental factors on episodes of mind-wandering. This suggests further investigations regarding the effects of environmental distractions.

To address this issue, Varao-Sousa et al. (2018c) investigated the effects of environmental factors on attention, mind-wandering, and memory. They observed higher rates of distraction outside the controlled lab setting. Consequently, participants in the outside condition performed worse in the memory test. These findings illustrate the expected negative effect of distraction on cognitive performance and decision-making. Li et al. (2018b) presented a model to account for the effects of distraction across various situations. This is critical as distraction can lead to more susceptibility to irrelevant information, thus negatively impacting decision-making.

Furthermore, Vredeveltdt and Perfect (2014b) explored different strategies that can help minimize and decrease distraction leading to better cognitive performance. Noise reduction is suggested as one of these strategies which can be beneficial for students to improve their concentration.

On the other hand, this issue needs to be further investigated as there is no definite answer to the question of whether mind-wandering will always have a positive effect during problem-solving. Studies such as Baird et al. (2012b) and Preiss et al. (2016), found mind-wandering to be helpful for creativity and problem-solving. It can be noted that mind-wandering can help individuals explore various solutions to a given problem, especially at the incubation stage. However, Rummel et al. (2020) questioned the previous theoretical claims regarding the positive effect of mind wandering throughout problem-solving. This was due to the reason that they found no significant difference in the performances of participants in either mindful or mindless conditions in incubation.

To conclude, our research can gain more insights into this topic and bridge this apparent gap by exploring more deeply how mind-wandering can impact problem-solving, especially route-planning tasks which require participants to concentrate and find optimal solutions. We will also examine whether this potential effect can be influenced by distracting noise, which resembles real-world daily situations; thus contributing to the existing studies in this field and making it practical for different organizations.

Methods

The experiment is going to be conducted in the following conditions: (1) The condition without any distracting background noise: Participants perform the route planning task in a quiet environment without any noise distractions. (2) The condition with distracting background noises (traffic noises): Participants perform the route planning task while exposed to simulated traffic noises in the background.

As mentioned above, participants in the experiment are assigned to either the "noises" or "no noises" condition using a between-subjects design. The amount of mind wandering during the problem and the performance on the route planning task are the dependent variables, and the independent variable is the presence of distraction noise.

Materials going to be used in the experiment include a desktop computer/ laptop, a route planning task (a type of problem-solving task), distracting background traffic noises, headphones, and a questionnaire at the end of the task (asking about participants' experiences and hearing abilities). To gauge the participant's current mental state and degree of mind wandering, pop-up questions will appear every two minutes during the route planning task, which requires participants to find the fastest route from point A to point B. Participants will be graded on their performance based on criteria such as overall travel time and costs.

Whether participants have hearing issues or noise sensitivity can be considered a confounding variable during the experiment as distracting background noises may not have enough desired effect on these participants. It will be mentioned in the research announcement that only students without any hearing issues or noise sensitivity can participate. One other potential confounding variable can be the stress levels of the participants as it can affect their mental situation and their performance on the test. Since noise can cause stress which can vary the performances systematically across the two conditions.

Statistical tests can be used to analyze the data and findings of our research. A t-test to compare the grades of the two conditions, namely the group with or without the background noise, giving an understanding of how noise distractions affect performance. Furthermore, with the mind wandering and noise variables dummy coded, a mixed linear regression analysis will be performed to investigate the association between noise, mind wandering, and grades. This will help us see which one can explain more variation in task performances and also control the confounding variable(s). Lastly, to find out if distraction noise causes mind-wandering, a straightforward correlation analysis will examine the association between noise distractions and mind-wandering.

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